

What the room does (right panel of Fig. 2)

How to read the map. Row = spot on the arc, column = frequency band. White: the spot reads what a circle predicts. Red: too loud. Blue: too quiet. Each microphone's own error is already taken out (left panel), so the colour is the room.

Deep blue at -72° and red at $+36^\circ/+54^\circ$, 500-1000 Hz. A sound wave that bounces off a nearby surface arrives a little later than the direct wave. If the extra path is half a wavelength the two cancel and the spot goes blue; if it is a whole wavelength they add and the spot goes red. The extra path depends on the spot, so neighbours can be opposite colours and one spot can flip colour from band to band. Wavelength here is 35-70 cm, so the surface is within about a metre.

The 200-250 Hz column: the blade tone. -36° is deep blue, $+36^\circ$ red. Wavelength about 1.4 m: only a large surface (wall, floor, table) reflects a wave that long. The blue cell moves with motor speed because the tone slides through the cancelling frequency, the textbook sign of a reflection.

Blue on one side, red on the other: the surface is off-centre. A surface symmetric around the propeller (the floor under a flat arc, say) gives every microphone the same extra path, so a whole row would be one colour. Here the colour changes around the arc, so the microphones are at different distances from the surface: it sits to one side. The colours do not say which side; blue is not "near" and red is not "far". The corner is the candidate because -72° is the spot closest to it.

White above 4 kHz. Short waves are absorbed by the foam and scattered by small objects, so only direct sound reaches the microphones. The arc itself is fine.

What we ruled out

The microphones. Because the microphones at $+72^\circ$ and $+54^\circ$ were swapped after prop10, and the arc was turned around for prop11-14, the fit can tell a bad microphone from a bad spot. Ten microphones agree within ± 1.8 dB in every band (left panel of Fig. 2). The one exception is **811-1892, which reads 2-6 dB too low from 125 Hz to 2.5 kHz**: its calibration file does not match the capsule. When it sat at $+54^\circ$ it made the room error look like 7 dB. Take it off the arc or measure it again. With it removed, the right panel does not change shape.

A labelling error. prop11–14 were measured with the arc turned around. Their patterns match the mirror image of the other runs (correlation +0.79 to +0.98). Plotted by label the dip appears on the other side; plotted by physical position they agree with the other eight runs. Turning the arc moves the labels, not the blue and red patches.

The propeller. With the arc flat and the axis vertical every microphone sees the propeller from the same angle, so the propeller cannot make one spot differ from another. Whatever is not white in Fig. 2 comes from outside the propeller.

Fig. 2 · Microphone error map (left) and room error map (right), same fit, same scale

Figure 2 consists of two heatmaps side-by-side. The left heatmap is titled 'Mic term M, dB re set average (row = its spot)' and the right is titled 'Room term P, dB from a circle at that spot'. Both heatmaps have a color scale on the right ranging from -6 (blue) to 6 (red). The x-axis for both is frequency in Hz (125 to 8000) and the y-axis is frequency in Hz (125 to 8000). The mic term heatmap shows a clear pattern of positive and negative values across the frequency spectrum, while the room term heatmap shows a more uniform distribution of values.

What to look for in the chamber

Three suspects, one per band in Fig. 2. Each is a size and a place, read off the map: a surface shows up once the wavelength is shorter than it (Rajmane and Baumann, DAGA 2016: a 35 cm plate from 1.25 kHz, a 10 cm plate from 3.15 kHz).

1. **200-250 Hz, blue at -36° .** Something about 1.5 m across, off to one side of the rig, with a bounced path 0.7 m longer than the direct one at -36° : a wall, a table top or the corner within about half a metre of that microphone.
2. **630-1000 Hz, blue at -72° .** Something about 0.4 m across near the -90° end of the arc on the corner side: the stand base, a bar of the arc frame, a cable tray.
3. **3150 Hz, scattered red and blue.** Something about 0.1 m across at the microphones themselves: clamps, mounts, the cable loop.

Next test (one afternoon)

1. Take 811-1892 off the arc.
2. Put a small loudspeaker on the stand instead of the propeller. Play short tone bursts at 250 and 800 Hz and record all microphones. The delay of the second arrival at each microphone, times 343 m/s, is the extra path length. Only one surface fits all eleven delays. With no airflow, anything left is the room.
3. Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
4. Cover or move that surface, then run scripts/arc_error_map.py again. Foam only helps at 3 kHz. At 250 Hz the only fix is distance.

How the numbers were made (not a sum, not an average)

Every level in the twelve runs (11 microphones $\times$ 19 third-octave bands $\times$ 12 runs) is written as three parts added together: **run gain** + **position error** + **microphone offset**. The three parts are found by least squares, i.e. the single set of values that fits all 12 $\times$ 11 levels best in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **Microphone offset** takes up each serial's own error after calibration. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

The turned-around runs are handled before the `arc`: for prop11-14 the label `+e` is entered as physical `-e`, so `"-72°"` always means the same spot in the room. The four microphone offsets and the eleven position errors can be told apart only because the same spot was covered by different microphones (the swap after prop10 and the four turned runs). Without that, a low microphone and a low spot would be indistinguishable. Spectra: Welch 4096-point Hann, UMIK-2 calibration as the app applies it, third-octave sums 125 Hz–8 kHz. Data: the last PWM-1900 capture in each key 2004_61n_unset_dpl-baseline-horizontal-prop7..prop18, 11 mics $\times$ 2.0 s; prop11-14 were measured with the arc turned around; prop1-6 and the 31 Aug keys are left out because they match neither orientation. Capture ids per key: `docs/analysis/arc/validation-data.json`. Code: `scripts/arc_error_map.py`, function `fit`.

Do the numbers add up?

Refitting from the data pack reproduces the published maps to 0.01 dB; each map sums to zero per band by construction. 132 measured cells per band are explained by 34 unknowns. 87% of all cells are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells (+11 dB at –36°/250 Hz in prop7; +6 to +7 dB at 0°/2–4 kHz in prop7–8) are single tone-band cells in the first two runs where the blade tone sits on a band edge; they do not follow a microphone or a spot and move the maps by under 1 dB. Limit: the 0° microphone (810–8897) never moved, so its M and the 0° P are known only as a sum.

Worked example · -72° spot at 800 Hz, two mics, two orientations

RUN	MIC THERE	MEASURED	GAIN	P	M	MODEL	LEFT
prop18	811-1896	59.1	62.3	-4.6	+2.5	60.2	-1.1
prop12 (180°)	810-8901	58.6	62.2	-4.6	+1.4	59.0	-0.4

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4;
prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.